Trajectory Tracking Control of Quadrotor Drones Based on Disturbance Observers

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***Abstract*—A composite control method is proposed by combining a disturbance observer with sliding mode control to address the uncertainties of parameters, models and external disturbances in the position and attitude control system for quadrotor unmanned aerial vehicles (UAVs). The observer estimates the equivalent disturbances in real-time, and a sliding mode controller is designed based on these estimates to track the desired signals. The designed controller not only ensures the system's asymptotic stability but also effectively suppresses the impact of internal and external disturbances. Finally, simulations verify the effectiveness of the proposed method.**

Keywords-Quadrotor drones; sliding mode control; disturbance observers; trajectory tracking

I. INTRODUCTION

Quadrotor drones, due to their outstanding flight characteristics and superior flexibility, have been applied in multiple fields such as search and rescue, logistics transportation, and aerial photography, showcasing their immense potential for application. However, the challenges remain significant in the development of high-precision control systems, especially when facing multiple disturbances such as external environmental conditions, and uncertainties in internal parameters and models.

In the realm of controller design for quadrotor drones, literature [1] proposed a method that combines PID with genetic algorithms, optimizing PID controller parameters to effectively reduce tracking errors. PID control performs well in controlling linear systems but is less suited for nonlinear dynamic systems and strongly coupled systems, such as quadrotors. Literature [2-4] demonstrated the use of active disturbance rejection control to overcome the challenges faced by traditional PID control in handling objects with high uncertainty. However, this method's parameter adjustment is relatively complex and not convenient for engineering applications. Sliding mode control, as a powerful nonlinear control strategy, has been applied to the control of quadrotor drones in literature [5-6]. These methods have achieved significant success in suppressing nonlinear dynamics and coupling effects, but the sliding mode controller can cause chattering phenomena when its output is large, and the control performance is not ideal under various disturbances.

Regarding the control strategy for disturbance attenuation, observers can estimate unknown disturbances and effectively compensate for them through a feedforward mechanism [7]. Literature [8] introduced a method that combines disturbance observers with optimal control techniques aimed at estimating and compensating for wind disturbances, achieving precise position and attitude control of the drone. Literature [9] addressed the control issues of quadrotor drones under mixed disturbances such as periodic disturbances and wind disturbances, proposing a combination scheme of disturbance observers (DO) and extended state observers (ESO), effectively suppressing these composite disturbances and achieving precise trajectory tracking. In literature [10-11], by combining NDO and other methods, the study focused on composite controllers with a hierarchical structure and established the stability of the closed-loop system. Disturbance observers not only achieve disturbance attenuation but are also easy to integrate with other traditional control methods.

Based on the analysis above, facing challenges such as parameter uncertainty, external time-varying disturbances, and uncertainties in the dynamic model, a composite control strategy that integrates Nonlinear Disturbance Observers (NDO) with Sliding Mode Control (SMC) is proposed to enhance the position tracking accuracy of quadrotors, thereby improving their trajectory tracking precision. On the one hand, the nonlinear disturbance observer estimates and mitigates the effects of both internal and external disturbances on the attitude loop through a feedforward approach; on the other hand, the disturbance observer compensates for the equivalent disturbances in the position loop in real-time, reducing their impact on position tracking. Finally, the stability of the closed-loop system is analyzed using the Lyapunov method, and anti-disturbance experiments are designed to verify the effectiveness of the proposed control method.

II. QUADROTOR DYNAMICS MODEL

Assuming that the quadrotor drone has a rigid body structure with its geometric center coinciding with the center of gravity, the coordinate system is illustrated in Figure 1, where represents the world-fixed frame, and represents the body-fixed frame.

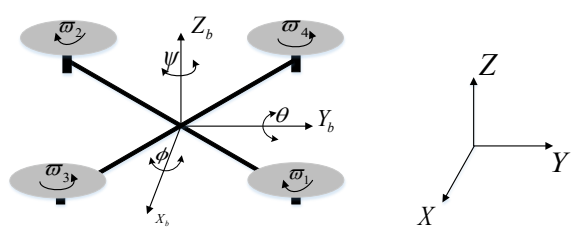

Figure 1. Schematic diagram of the quadrotor drone structure.

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The rotational speeds of the propellers are defined as ϖ_1, ϖ_2, ϖ_3, and ϖ_4, respectively, with the total lift force along the Z_b axis denoted as f, and the rotational torques about the X_b, Y_b, and Z_b axes represented by τ_ϕ, τ_θ, and τ_ψ, respectively. The relationship between the lift force f, rotational torques τ, and the propeller speeds is as follows:

$$\begin{bmatrix} f \\ \tau_\phi \\ \tau_\theta \\ \tau_\psi \end{bmatrix} = \begin{bmatrix} c_T & c_T & c_T & c_T \\ -\frac{\sqrt{2}}{2}dc_T & \frac{\sqrt{2}}{2}dc_T & \frac{\sqrt{2}}{2}dc_T & -\frac{\sqrt{2}}{2}dc_T \\ \frac{\sqrt{2}}{2}dc_T & -\frac{\sqrt{2}}{2}dc_T & \frac{\sqrt{2}}{2}dc_T & -\frac{\sqrt{2}}{2}dc_T \\ c_M & c_M & c_M & c_M \end{bmatrix} \begin{bmatrix} \varpi_1^2 \\ \varpi_2^2 \\ \varpi_3^2 \\ \varpi_4^2 \end{bmatrix} \tag{1}$$

where d represents the distance from the drone's center to the motor's axis of rotation, with c_T denoting the propeller thrust coefficient and c_M the propeller torque coefficient.

The positional and attitudinal dynamics of the quadrotor drone can be represented as follows:

$$\begin{cases} m\ddot{x} = F_x + k_t\dot{x} + d_x \\ m\ddot{y} = F_y + k_t\dot{y} + d_y \\ m\ddot{z} = F_z - mg + k_t\dot{z} + d_z \\ \tilde{I}_x\ddot{\phi} = (\tilde{I}_y - \tilde{I}_z)\dot{\theta}\dot{\psi} + \tau_\phi - J_{rp}\Omega\dot{\theta} + d_\phi \\ \tilde{I}_y\ddot{\theta} = (\tilde{I}_z - \tilde{I}_x)\dot{\phi}\dot{\psi} + \tau_\theta + J_{rp}\Omega\dot{\phi} + d_\theta \\ \tilde{I}_z\ddot{\psi} = (\tilde{I}_x - \tilde{I}_y)\dot{\phi}\dot{\theta} + \tau_\psi + d_\phi \end{cases} \tag{2}$$

where m represents the total mass of the drone and its payload, x, y, z represent the positions of the drone along three axes in world-fixed frame, respectively, and F_x, F_y, F_z are the equivalent control forces in the three axes of the position loop, respectively. g is the gravity acceleration, k_t is the drag coefficient, and d_x, d_y, d_z denote the external disturbance forces encountered in the three axes of the position loop, respectively. $\tilde{I}_x, \tilde{I}_y, \tilde{I}_z$ represent the rotational inertias of the quadrotor, with ϕ, θ, ψ corresponding to the roll, pitch, and yaw angles, respectively. J_{rp} signifies the rotational inertia of the propellers, and $\Omega = -\varpi_1 - \varpi_2 + \varpi_3 + \varpi_4$. d_ϕ, d_θ, d_ψ represent the external disturbance torques affecting the attitudinal loop.

The relationship between the equivalent control force F and the total lift force f of the drone can be expressed as follows:

$$F = \begin{bmatrix} F_x \\ F_y \\ F_z \end{bmatrix} = f \begin{bmatrix} \cos\psi\sin\theta\cos\phi + \sin\psi\sin\phi \\ \sin\psi\sin\theta\cos\phi - \cos\psi\sin\phi \\ \cos\phi\cos\theta \end{bmatrix} \tag{3}$$

During flight, drones may carry heavy loads, which, when not perfectly aligned with the drone's geometric center, can affect its rotational inertia. The quadrotor's rotational inertia $\tilde{I}_i$ is a combination of nominal (I_i) and uncertain (ΔI_i) parts as

$$\tilde{I}_i = I_i + \Delta I_i (i = x, y, z) \tag{4}$$

The position and attitude kinematics of the quadrotor drone can be reformulated as follows:

$$\begin{cases} \ddot{x} = \frac{1}{m}F_x + \tilde{d}_x \\ \ddot{y} = \frac{1}{m}F_y + \tilde{d}_y \\ \ddot{z} = \frac{1}{m}F_z + \tilde{d}_z \\ \ddot{\phi} = \frac{I_y - I_z}{I_x}\dot{\theta}\dot{\psi} + \frac{1}{I_x}\tau_\phi + \frac{1}{I_x}\tilde{d}_\phi \\ \ddot{\theta} = \frac{I_z - I_x}{I_y}\dot{\phi}\dot{\psi} + \frac{1}{I_y}\tau_\theta + \frac{1}{I_y}\tilde{d}_\theta \\ \ddot{\psi} = \frac{I_x - I_y}{I_z}\dot{\phi}\dot{\theta} + \frac{1}{I_z}\tau_\psi + \frac{1}{I_z}\tilde{d}_\phi \end{cases} \tag{5}$$

And

$$\begin{cases} \tilde{d}_x = \frac{k_t}{m}\dot{x} + \frac{1}{m}d_x \\ \tilde{d}_y = \frac{k_t}{m}\dot{y} + \frac{1}{m}d_y \\ \tilde{d}_z = \frac{k_t}{m}\dot{z} + \frac{1}{m}d_z \\ \tilde{d}_\phi = -\Delta I_x\dot{\phi} + \frac{\Delta I_y - \Delta I_z}{\Delta I_x}\dot{\theta}\dot{\psi} - J_{rp}\Omega\dot{\theta} + d_\phi \\ \tilde{d}_\theta = -\Delta I_y\dot{\theta} + \frac{\Delta I_z - \Delta I_x}{\Delta I_y}\dot{\phi}\dot{\psi} + J_{rp}\Omega\dot{\phi} + d_\theta \\ \tilde{d}_\psi = -\Delta I_z\dot{\psi} + \frac{\Delta I_x - \Delta I_y}{\Delta I_z}\dot{\phi}\dot{\theta} + d_\psi \end{cases} \tag{6}$$

where $\tilde{d}_x, \tilde{d}_y, \tilde{d}_z$ represent the equivalent disturbances in the position loop, and $\tilde{d}_\phi, \tilde{d}_\theta, \tilde{d}_\psi$ denote the equivalent disturbances in the attitude loop.

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Assumption 1. The roll, pitch, and yaw angles of the drone all satisfy $\phi \in (-\frac{\pi}{2}, \frac{\pi}{2})$, $\theta \in (-\frac{\pi}{2}, \frac{\pi}{2})$, $\psi \in (-\frac{\pi}{2}, \frac{\pi}{2})$.

Assumption 2. The equivalent disturbance $\tilde{d}_i$ is continuous and bounded, and its derivative is also bounded and there exist two constants $d_{\max}, \bar{d}$ such that $\left\|\tilde{d}_i\right\| \le d_{\max}$, $\left\|\dot{\tilde{d}}_i\right\| \le \bar{d}$. $(i = x, y, z, \phi, \theta, \psi)$

III. Design of Control Systems

A. Design of the Observer

The designed nonlinear disturbance observer (NDO) and disturbance observer (DO) in this section are aimed at respectively estimating unknown disturbances in the attitude and position loops for compensation.

1) Design of the Nonlinear Disturbance Observer

The attitude dynamics of system (5) are reformulated as follows:

$$\ddot{\eta} = f + B_a \tau + B_a \tilde{d} \tag{7}$$

where $\eta = \begin{bmatrix} \phi & \theta & \psi \end{bmatrix}^T$ denotes the system state variable, $f = \begin{bmatrix} \frac{I_y - I_z}{I_x}\dot{\theta}\dot{\psi} & \frac{I_z - I_x}{I_y}\dot{\phi}\dot{\psi} & \frac{I_x - I_y}{I_z}\dot{\phi}\dot{\theta} \end{bmatrix}^T$, $B_a = diag(\frac{1}{I_x} \quad \frac{1}{I_y} \quad \frac{1}{I_z})$, $\tau = \begin{bmatrix} \tau_\phi & \tau_\theta & \tau_\psi \end{bmatrix}^T$, $\tilde{d}_a = \begin{bmatrix} \tilde{d}_\phi & \tilde{d}_\theta & \tilde{d}_\psi \end{bmatrix}^T$.

To estimate the equivalent disturbance $\tilde{d}_a$, a nonlinear disturbance observer is selected as follows [12]:

$$\begin{cases} \dot{z}_a = -l_a(\dot{\eta})\{B_a[\lambda_a(\dot{\eta}) + z_a] + (f + B\tau)\} \\ \hat{\tilde{d}}_a = \lambda_a(\dot{\eta}) + z_a \end{cases} \tag{8}$$

where $\hat{\tilde{d}}_a$ denotes the estimated disturbance, z_a represents the observer state variable, $l_a(\dot{\eta})$ is the gain matrix of the disturbance observer, and $\lambda_a(\dot{\eta})$ is a nonlinear function, with their relationship defined as follows:

$$l_a(\dot{\eta}) = \frac{\partial \lambda_a(\dot{\eta})}{\partial \dot{\eta}} \tag{9}$$

Theorem 1: By selecting the gain $l_a(\dot{\eta}) = PI_{3\times3}$ of the nonlinear disturbance observer and the nonlinear function $\lambda_a(\dot{\eta}) = P\dot{\eta}$, where $P = diag(P_1 \quad P_2 \quad P_3)$, $I_{3\times3}$ is the identity matrix and P_1, P_2, P_3 are positive, it is guaranteed that the steady-state error of the disturbance estimation is bounded.

Proof: Define the attitude loop disturbance estimation error

$$e_a = \tilde{d}_a - \hat{\tilde{d}}_a \tag{10}$$

Upon differentiation, it yields:

$$\dot{e}_a = \dot{\tilde{d}}_a - \dot{\hat{\tilde{d}}}_a = \dot{\tilde{d}} - \dot{z}_a - \dot{\lambda}_a = \dot{\tilde{d}}_a - l_a(x)B_a e_a \tag{11}$$

Define the Lyapunov function as:

$$V_1 = \frac{1}{2} e_a{}^T e_a \tag{12}$$

Differentiating, one obtains:

$$\begin{aligned} \dot{V}_1 &= e_a{}^T \dot{e}_a = e_a{}^T [\dot{\tilde{d}}_a - l_a(x)B_a e_a] \\ &= -e_a{}^T l_a(x) B_a e_a + e_a{}^T \dot{\tilde{d}}_a \\ &\le -p_a \mu_a e_a{}^T e_a + e_a{}^T \dot{\tilde{d}}_a \\ &\le -p_a \mu_a \left\|e_a\right\|^2 + \left\|e_a\right\| \bar{d} \end{aligned} \tag{13}$$

where p_a represents the minimum eigenvalue of matrix P, μ_a signifies the minimum eigenvalue of matrix B_a.

Since matrices B_a and P are positive definite, it follows that $\mu_a > 0$ and $p_a > 0$. When $\left\|e_a\right\| > \bar{d} / p_a \mu_a$ is chosen, $\dot{V}_1$ becomes negative definite. Hence, an upper bound for the steady-state estimation error can be derived as:

$$\left\|e_a\right\| < \frac{\bar{d}}{p_a \mu_a} \tag{14}$$

Therefore, it is reasonable to choose the disturbance observer gain to keep the estimation error within a certain range. At this point, the disturbance observer estimates the equivalent disturbance $\tilde{d}_a$ with a certain level of accuracy.

2) Design of the Disturbance Observer

To estimate the equivalent disturbance $\tilde{d}_a$, the disturbance observers for channels X, Y, Z are designed as follows $(j = x, y, z)$:

$$\begin{cases} \dot{\tilde{d}}_j = -k_d \rho_j - \varepsilon \rho_j^{\frac{p_0}{q_0}} - Tsat(\rho_j) \\ \rho_j = z_j - v_j \\ \dot{z}_j = -k_p \rho_j - \varepsilon \rho_j^{\frac{p_0}{q_0}} - Tsat(\rho_j) + \frac{1}{m} F_j \end{cases} \tag{15}$$

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where $\hat{d}_j$ represents the disturbance estimation of position loop, $k_d, \varepsilon, p_0, q_0, T$ are observer gains, and ρ_j, z_j denote the internal states of the observer. The saturation function S is defined as follows:

$$sat(x) = \begin{cases} 1, x > k \\ \dfrac{x}{k}, -k \le x \le k \\ -1, x < -k \end{cases} \tag{16}$$

Theorem 2: By selecting appropriate disturbance observer gain $k_d, \varepsilon, p_0, q_0$ all positive, and satisfying $p_0 < q_0, T > d_{\max}$. It ensures that the disturbance observer can track the equivalent disturbance $\tilde{d}_x, \tilde{d}_y, \tilde{d}_z$ effectively.

Proof: Define the position loop disturbance estimation error $(j = x, y, z)$:

$$\begin{aligned} e_j &= \hat{\tilde{d}}_j - \tilde{d}_j = -k_d \rho_j - \varepsilon \rho_j^{\frac{p_0}{q_0}} - Tsat(\rho_j) - \tilde{d}_j \\ &= -k_d \rho_j - \varepsilon \rho_j^{\frac{p_0}{q_0}} - Tsat(\rho_j) + \frac{1}{m} F_j - \dot{v}_j \\ &= \dot{z}_j - \dot{v}_j \\ &= \dot{\rho}_j \end{aligned} \tag{17}$$

Define the Lyapunov function as:

$$V_2 = \frac{1}{2} \rho_j^2 \tag{18}$$

Differentiating, one obtains:

$$\begin{aligned} \dot{V}_2 &= \rho_j \dot{\rho}_j = \rho_j [-k_d \rho_j - \varepsilon \rho_j^{\frac{p_0}{q_0}} - Tsat(\rho_j) - \tilde{d}_j] \\ &\le -k_d \rho_j^2 - \varepsilon \rho_j^{\frac{p_0+q_0}{q_0}} - T|\rho_j| + |\tilde{d}_j||\rho_j| \\ &\le -k_d \rho_j^2 - \varepsilon \rho_j^{\frac{p_0+q_0}{q_0}} \end{aligned} \tag{19}$$

From equation (24), it follows that:

$$\dot{V}_2 \le -2k_d V_2 - 2^{\frac{p_0+q_0}{2q_0}} \varepsilon V_2^{\frac{p_0+q_0}{2q_0}} \tag{20}$$

Based on [13], it is evident that variable ρ_j converges to 0 within a finite time t_d, thereby ensuring the convergence of $e_j = \dot{\rho}_j$ within a finite time.

B. Design of the controller

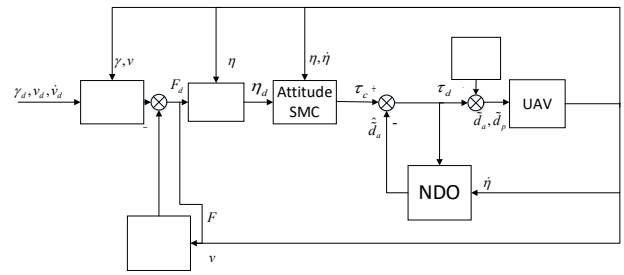

Figure 2. Scheme diagram of the quadrotor drone structure.

The specific structure of the sliding mode control with composite disturbance observer adopted in this paper is shown in Figure 2. The sliding mode controller mainly controls the attitude and position so that the desired value can be reached quickly and makes the control system robust, while the disturbance observer is used to estimate and compensate for the disturbance, which is used to improve the control accuracy of the system.

1) Design of attitude controller

The attitude angle tracking error is defined as $e_\eta = \eta - \eta_d$, where $\eta_d = [\phi_d \quad \theta_d \quad \psi_d]^T$ represents the desired attitude angle.

For system (5), the sliding mode surface is chosen in the following form:

$$s_a = \begin{bmatrix} s_\phi \\ s_\theta \\ s_\psi \end{bmatrix} = \dot{e}_\eta + \alpha_a e_\eta + \beta_a e_\eta^{\frac{q_1}{p_1}} \tag{21}$$

where $\alpha_a = diag(\alpha_1 \quad \alpha_2 \quad \alpha_3)$ and $\beta_a = diag(\beta_1 \quad \beta_2 \quad \beta_3)$ are positive definite matrices, and $0 < q_1 < p_1$.

Taking the derivative of the sliding mode surface yields:

$$\dot{s}_a = \ddot{e}_\eta + \alpha_a \dot{e}_\eta + \frac{q_1}{p_1} \beta_a e_\eta^{\frac{q_1}{p_1}-1} \dot{e}_\eta \tag{22}$$

the convergence rate u_a is selected as follows:

$$u_a = \begin{bmatrix} -\rho_{1\phi} sat(s_\phi) - \rho_{2\phi} s_\phi \\ -\rho_{1\theta} sat(s_\theta) - \rho_{2\theta} s_\theta \\ -\rho_{1\psi} sat(s_\psi) - \rho_{2\psi} s_\psi \end{bmatrix} \tag{23}$$

For attitude motion systems, a sliding mode attitude controller can be defined as follows:

$$\tau_c = B_a^{-1}(-f + \ddot{\eta}_d - \alpha_a \dot{e}_\eta - \frac{q_1}{p_1} \beta_a e_\eta^{\frac{q_1}{p_1}-1} \dot{e}_\eta + u_a) \tag{24}$$

The composite sliding mode control law based on nonlinear disturbance observer is defined as follows:

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$$\tau_d = B_a^{-1}(-f + \ddot{\eta}_d - \alpha_a \dot{e}_\eta - \frac{q_1}{p_1}\beta_a e_\eta^{\frac{q_1}{p_1}-1}\dot{e}_\eta + u_a) - \hat{\tilde{d}}_a \quad (25)$$

The desired attitude angular velocity $\dot{\eta}_d, \ddot{\eta}_d$ are set to zero, which is applicable to quadrotor UAVs operating within a small angle range.

Theorem 3: For system (5), when the control law is chosen as (25) and gains satisfy $\min\{\rho_{1\phi} \quad \rho_{1\theta} \quad \rho_{1\psi}\} > \mu\bar{d} / p_a\mu_a$ and $\rho_{2\phi}, \rho_{2\theta}, \rho_{2\psi}$ both are positive, the system is asymptotically stable.

Proof: Substituting equation (25) into equation (5) yields:

$$\dot{s}_a = u_a + B_a e_a \quad (26)$$

Define the Lyapunov function as:

$$V_a = \frac{1}{2} s_a^T s_a \quad (27)$$

Differentiating, one obtains:

$$\begin{aligned}
\dot{V}_a &= s_a^T \dot{s}_a = s_a^T (u_a + B_a e_a) \\
&= -\rho_{1\phi} s_\phi sat(s_\phi) - \rho_{2\phi} s_\phi^2 + \frac{1}{I_x} s_\phi e_\phi \\
&\quad - \rho_{1\theta} s_\theta sat(s_\theta) - \rho_{2\theta} s_\theta^2 + \frac{1}{I_y} s_\theta e_\theta \\
&\quad - \rho_{1\psi} s_\psi sat(s_\psi) - \rho_{2\psi} s_\psi^2 + \frac{1}{I_z} s_\psi e_\psi \quad (28) \\
&\le -\rho_{1\phi}|s_\phi| - \rho_{2\phi} s_\phi^2 + \frac{1}{I_x}|s_\phi||e_\phi| \\
&\quad - \rho_{1\theta}|s_\theta| - \rho_{2\theta} s_\theta^2 + \frac{1}{I_y}|s_\theta||e_\theta| \\
&\quad - \rho_{1\psi}|s_\psi| - \rho_{2\psi} s_\psi^2 + \frac{1}{I_z}|s_\psi||e_\psi|
\end{aligned}$$

where $e_a = \begin{bmatrix} e_\phi & e_\theta & e_\psi \end{bmatrix}^T$.

Let μ be defined as the maximum eigenvalue of matrix B_a. From equation (28), it is evident that when $\min\{\rho_{1\phi} \quad \rho_{1\theta} \quad \rho_{1\psi}\} > \mu\bar{d} / p_a\mu_a$ and $\rho_{2\phi}, \rho_{2\theta}, \rho_{2\psi}$ are all positive, $\dot{V} < 0$. Therefore, the closed-loop system is asymptotically stable.

2) Design of attitude controller

Define the position tracking error $e_\gamma = \gamma - \gamma_d$, velocity tracking error $e_v = v - v_d$, where $\gamma = [x \quad y \quad z]^T$ and $v = \begin{bmatrix} \dot{x} & \dot{y} & \dot{z} \end{bmatrix}^T$ represent the position and velocity vectors in the drone's world-fixed frame, and γ_d, v_d are the desired position and desired velocity, respectively.

For the position motion system (5), we select the following form of sliding mode surface:

$$s_p = \begin{bmatrix} s_x \\ s_y \\ s_z \end{bmatrix} = \dot{e}_\gamma + \alpha_p e_\gamma + \beta_p e_\gamma^{\frac{q_2}{p_2}} \quad (29)$$

where $\alpha_p = diag(\alpha_4 \quad \alpha_5 \quad \alpha_6)$ and $\beta_a = diag(\beta_4 \quad \beta_5 \quad \beta_6)$ are positive definite matrices, and $0 < q_2 < p_2$.

Taking the derivative of the sliding mode surface yields:

$$\dot{s}_p = \ddot{e}_\gamma + \alpha_p \dot{e}_\gamma + \frac{q_2}{p_2}\beta_p e_\gamma^{\frac{q_2}{p_2}-1}\dot{e}_\gamma \quad (30)$$

the convergence rate u_p is selected as follows:

$$u_p = \begin{bmatrix} -\rho_{1x} sat(s_x) - \rho_{2x} s_x \\ -\rho_{1y} sat(s_y) - \rho_{2y} s_y \\ -\rho_{1z} sat(s_z) - \rho_{2z} s_z \end{bmatrix} \quad (31)$$

For position motion systems, a sliding mode attitude controller can be defined as follows:

$$F_c = m\dot{v}_d - m\alpha_p e_v - m\frac{q_2}{p_2}\beta_p e_\gamma^{\frac{q_2}{p_2}-1}\dot{e}_\gamma + mu_p \quad (32)$$

Therefore, the composite sliding mode position control law based on disturbance observer is:

$$F_d = m\dot{v}_d - m\alpha_p e_v - m\frac{q}{p}\beta_p e_\gamma^{\frac{q}{p}-1}\dot{e}_\gamma + mu_p - m\hat{\tilde{d}}_p \quad (33)$$

where $\hat{\tilde{d}}_p = \begin{bmatrix} \hat{d}_x & \hat{d}_y & \hat{d}_z \end{bmatrix}^T$ represents the estimated disturbance by the observer.

Theorem 4: For system (7), when the control law is selected as (32) and the gains satisfy $\min(\rho_{1x} \quad \rho_{1y} \quad \rho_{1z}) > \max(|\dot{\rho}_x| \quad |\dot{\rho}_y| \quad |\dot{\rho}_z|)$, and $\rho_{2x}, \rho_{2y}, \rho_{2z}$ are all positive, the system is asymptotically stable.

proof: substituting equations (17), (31), and (33) into equation (5) yields:

$$\dot{s}_p = \begin{bmatrix} \dot{s}_x \\ \dot{s}_y \\ \dot{s}_z \end{bmatrix} = \begin{bmatrix} -\rho_{1x} sat(s_x) - \rho_{2x} s_x - \dot{\rho}_x \\ -\rho_{1y} sat(s_y) - \rho_{2y} s_y - \dot{\rho}_y \\ -\rho_{1z} sat(s_z) - \rho_{2z} s_z - \dot{\rho}_z \end{bmatrix} \quad (34)$$

Define the Lyapunov function as:

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$$V_p = \frac{1}{2} s_p{}^T s_p \tag{35}$$

Differentiating, one obtains:

$$\begin{aligned}\dot{V}_p &= s_p{}^T \dot{s}_p \\ &= -\rho_{1x} s_x sat(s_x) - \rho_{2x} s_x^2 - s_x \dot{\rho}_x \\ &\quad - \rho_{1y} s_y sat(s_y) - \rho_{2y} s_y^2 - s_y \dot{\rho}_y \\ &\quad - \rho_{1z} s_z sat(s_z) - \rho_{2z} s_z^2 - s_z \dot{\rho}_z \\ &\le -\rho_{1x} |s_x| - \rho_{2x} s_x^2 + |s_x||\dot{\rho}_x| \\ &\quad - \rho_{1y} |s_y| - \rho_{2y} s_y^2 + |s_y||\dot{\rho}_y| \\ &\quad - \rho_{1z} |s_z| - \rho_{2z} s_z^2 + |s_z||\dot{\rho}_z|\end{aligned} \tag{36}$$

From equation (36), it can be observed that when $\min(\rho_{1x} \quad \rho_{1y} \quad \rho_{1z}) > \max(|\dot{\rho}_x| \quad |\dot{\rho}_y| \quad |\dot{\rho}_z|)$ and $\rho_{2x}, \rho_{2y}, \rho_{2z}$ are all positive, $\dot{V}_p < 0$,thus indicating that the closed-loop system is asymptotically stable.

3) Attitude estimation

According to (3) and (33), the desired input for the attitude loop is driven by

$$\begin{cases} f = \dfrac{F_z}{\cos\phi \cos\theta} \\ \phi_d = \arctan(\cos\theta \dfrac{F_x \sin\psi - F_y \cos\psi}{F_z}) \\ \theta_d = \arctan(\dfrac{F_x \cos\psi + F_y \sin\psi}{F_z}) \end{cases} \tag{37}$$

where ψ_d set to zero in the paper.

IV. Simulation Results

In this section, the effectiveness of the proposed composite control method is validated through MATLAB/Simulink simulations. Comparative tests with and without disturbances observer in a quadrotor subjected to a combination of internal and external disturbances.

The structural parameters of the quadcopter attitude system are set as follows:

$$m = 1.5\text{kg}, I_x = 2.103\times10^{-2} kg\cdot m^2,$$
$$I_y = 2.163\times10^{-2} kg\cdot m^2, I_z = 3.942\times10^{-2} kg\cdot m^2,$$
$$d = 0.187m, c_T = 0.938\times10^{-5} N/rad/s^2,$$
$$c_M = 1.479\times10^{-7} N\cdot m/(rad/s)^2, J_{rp} = 8.19\times10^{-5} kg/m^2.$$

The observer parameters are:

$$p = diag\{5 \quad 5 \quad 3\}\ k_d = 10, \varepsilon = 10, p_0 = 2, q_0 = 3,$$
$$T = 80.$$

The sliding mode controller parameters are:

$$\rho_{1\phi} = 30,\ \rho_{1\theta} = 50,\ \rho_{1\psi} = 30,\ \rho_{2\phi} = 20,\ \rho_{2\theta} = 20,$$
$$\rho_{2\psi} = 20.\ \rho_{1x} = 10,\ \rho_{1y} = 10,\ \rho_{1z} = 10,\ \rho_{2x} = 8,$$
$$\rho_{2y} = 8,\ \rho_{2z} = 8.$$

The initial values of position and velocity for the quadrotor UAV are set to $\gamma(0)=[0.2 \quad 0.2 \quad 0]^T$ (m) and $v(0)=[0 \quad 0 \quad 0]^T$ (m/s) , respectively. To validate the effectiveness of the composite controller, we assume uncertain parameters ΔI_i to be 20% of the nominal system parameters I_i. $(i = \phi, \theta, \psi)$

The desired position and velocity for the quadrotor UAV are set to:

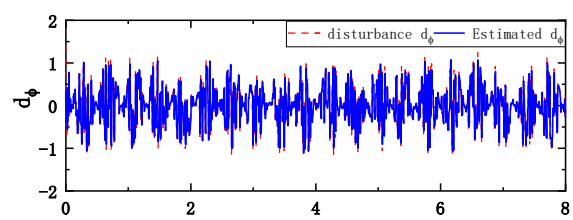
$$\gamma_d(t)=[1.8\sin t \quad 1.8\cos t \quad 0.5]^T$$
$$v_d(t)=[0.9\cos t \quad -0.9\sin t \quad 0.5]^T$$

The external disturbances in the position and attitude loops of the quadrotor UAV are respectively designated as:

$$d_p(t) = \begin{bmatrix} d_x \\ d_y \\ d_z \end{bmatrix} = \begin{bmatrix} 2.9\mathrm{R}_1 \cos(1.5t + \frac{\pi}{4}) \\ 1.8\mathrm{R}_2 \cos(1.2t + \frac{\pi}{3}) \\ 1.4\mathrm{R}_3 \cos(t + \frac{\pi}{6}) \end{bmatrix}$$

$$d_a(t) = \begin{bmatrix} d_\phi \\ d_\theta \\ d_\psi \end{bmatrix} = \begin{bmatrix} 0.9\mathrm{R}_4 \cos(5t + \frac{\pi}{4}) \\ 0.8\mathrm{R}_5 \cos(6t + \frac{\pi}{3}) \\ 0.4\mathrm{R}_6 \cos(5t + \frac{\pi}{6}) \end{bmatrix}$$

Where R_1 to R_6 are random numbers between -1 and 1.



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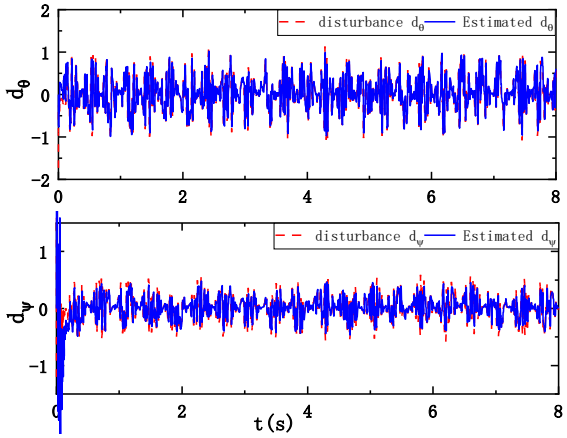

Figure 3. Comparison of disturbance in attitude loop.

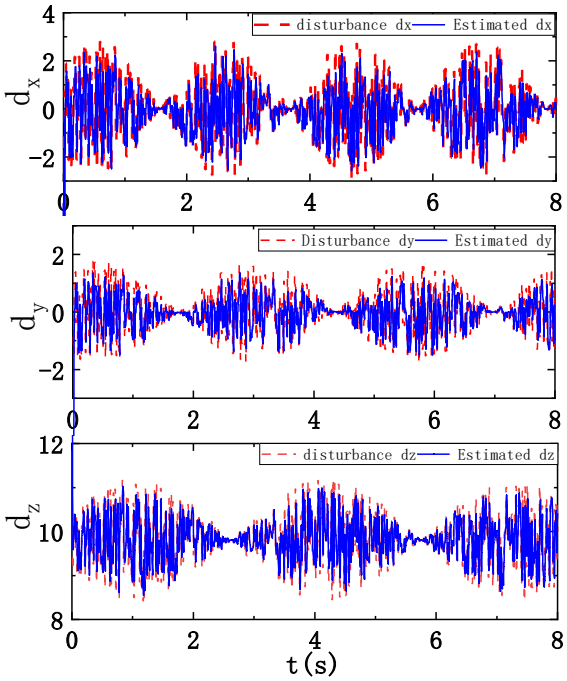

Figure 4. Comparison of disturbance in position loop.

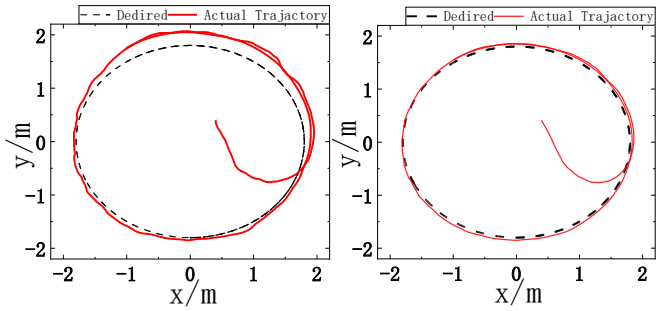

Figure 5. Comparison of trajectory tracking performance: the left image without any observer, and the right image with two observers.

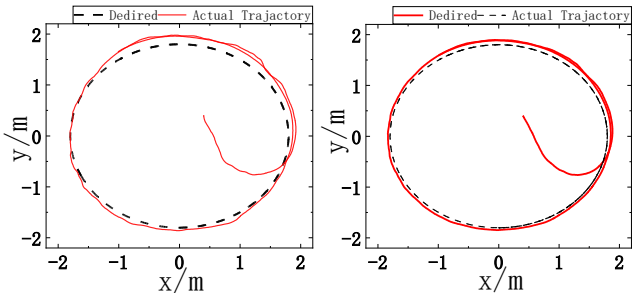

Figure 6. Comparison of trajectory tracking performance: the left image with only NDO, and the right image with only DO.

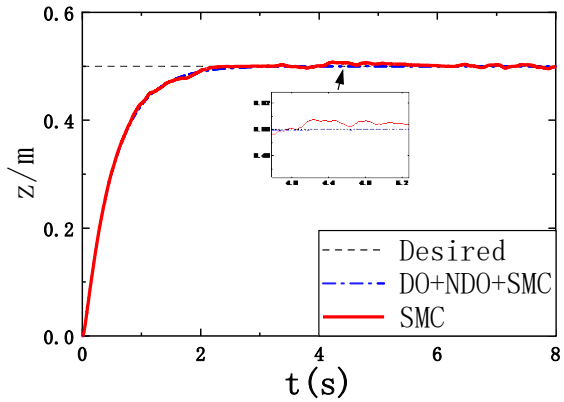

Figure 7. Quadrotor z-direction time response.

Figures 3 and 4 show the estimation of the equivalent disturbance by the nonlinear disturbance observer of the attitude loop and the disturbance observer of the position loop, respectively, from which it can be seen that both observers can estimate the disturbance effectively. From Figures 5 and 6, it can be seen that in the absence of an observer, the maximum error between the position of the quadcopter and the desired position is 0.23m, and there is a significant error in trajectory tracking compared to the desired trajectory. When a nonlinear disturbance observer is incorporated into the attitude loop, the position error of the quadcopter decreases to 0.19m, and the influence of disturbances on the UAV's attitude angles is reduced, leading to an improvement in trajectory tracking. Introducing a disturbance observer into the position loop while the attitude loop does not include a nonlinear disturbance observer results in a position error of 0.11m for the quadcopter, indicating a noticeable attenuation of disturbances in the position loop by the observer. Finally, with both position and attitude loops equipped with observers, the position error decreases to 0.06m, significantly enhancing the control performance. Furthermore, from Figure 7, it can be observed that compared to relying solely on the sliding mode controller and compensating for disturbances with two observers, the final steady-state error is reduced from 0.015m to 0.003m.

V. Conclusion

This study proposes a hybrid control strategy integrating disturbance observer and sliding mode control to enhance the

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robustness of quadrotor UAV trajectory control system against external disturbances. In this control strategy, a nonlinear disturbance observer is designed to estimate all unknown disturbances in the attitude control loop, while another disturbance observer is employed to estimate unknown disturbances in the position loop, effectively compensating the sliding mode controller via feedforward mechanism. The developed control scheme ensures smooth execution of attitude control tasks in diverse disturbance environments. Simulation results validate the effective disturbance estimation capability of the proposed disturbance observers and demonstrate significant reduction of disturbance effects on quadrotor UAV control system performance through feedforward compensation mechanism. The proposed hybrid controller effectively enhances the robustness and trajectory tracking precision of quadrotor UAVs.

ACKNOWLEDGMENT

The preferred spelling of the word "acknowledgment" in America is without an "e" after the "g". Avoid the stilted expression, "One of us (R. B. G.) thanks . . ." Instead, try "R. B. G. thanks". Put sponsor acknowledgments in the unnumbered footnote on the first page.

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